

The Continuous and the Discrete:

An Analytic Resolution of $G^* = \sqrt{2\pi} \cdot \theta_3(e^{-\pi})^2$

There exist rare identities in mathematics that serve as crossroads for entirely disparate fields. The expression $G^* = \sqrt{2\pi} \cdot \theta_3(e^{-\pi})^2$ is one such artifact, sitting at the deep intersection of modular forms, elliptic integrals, continuous probability, and discrete geometry. By unpacking this identity step-by-step, we reveal its exact closed-form value, its algebraic connection to historical constants, and its profound geometric interpretation as a bridge between continuous Euclidean space and the discrete integer lattice.

1. Evaluating the Jacobi Theta Function

The core engine of this identity is the Jacobi theta function $\theta_3(q)$, which is defined as the infinite sum of a Gaussian sequence over the integers:

$$\theta_3(q) = \sum_{n=-\infty}^{\infty} q^{n^2} \quad (1)$$

When we evaluate this at $q = e^{-\pi}$, we are probing the highly symmetric “self-dual” point in the complex upper half-plane ($\tau = i$). Because the Gaussian distribution is an eigenfunction of the Fourier transform, evaluating the discrete sum at this precise resonant parameter yields a known exact analytic value governed by the Euler gamma function (Γ):

$$\theta_3(e^{-\pi}) = \frac{\pi^{1/4}}{\Gamma(3/4)} \quad (2)$$

To render this into a more foundational form, we apply Euler’s reflection formula, $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$. Setting $z = 1/4$:

$$\begin{aligned} \Gamma(1/4)\Gamma(3/4) &= \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2} \\ \implies \Gamma(3/4) &= \frac{\pi\sqrt{2}}{\Gamma(1/4)} \end{aligned} \quad (3)$$

Substituting this reflection back into our evaluated theta function yields the exact amplitude of the one-dimensional discrete Gaussian:

$$\theta_3(e^{-\pi}) = \frac{\pi^{1/4}\Gamma(1/4)}{\pi\sqrt{2}} = \frac{\Gamma(1/4)}{\sqrt{2}\pi^{3/4}} \quad (4)$$

2. Squaring and the Continuous Measure

Next, the identity squares this discrete theta value. Geometrically, squaring the one-dimensional theta function gives us the partition function of a Gaussian over a two-dimensional square integer lattice ($\mathbb{Z}^2$):

$$\theta_3(e^{-\pi})^2 = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} e^{-\pi(m^2+n^2)} = \frac{\Gamma(1/4)^2}{2\pi^{3/2}} \quad (5)$$

Finally, we multiply this lattice sum by the continuous amplitude $\sqrt{2\pi}$, as prescribed by the identity, to uncover the exact closed algebraic form of G^* :

$$\begin{aligned} G^* &= \sqrt{2\pi} \cdot \frac{\Gamma(1/4)^2}{2\pi^{3/2}} \\ &= \frac{\sqrt{2} \pi^{1/2} \Gamma(1/4)^2}{2\pi^{3/2}} \end{aligned} \quad (6)$$

Simplifying the transcendental constants ($\pi^{1/2}/\pi^{3/2} = \pi^{-1}$) yields the true, fundamental nature of the expression:

$$\boxed{G^* = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}} \approx 2.9586751 \dots} \quad (7)$$

3. A Nexus of Nineteenth-Century Mathematics

Because G^* is driven fundamentally by the square of the gamma function evaluated at one-quarter, $\Gamma(1/4)^2$, it acts as a mathematical hub that elegantly connects several legendary nineteenth-century constants:

- **Gauss's Constant (G):** In 1799, Carl Friedrich Gauss discovered that the reciprocal of the arithmetic-geometric mean (agm) of 1 and $\sqrt{2}$ yields a fundamental constant of nature:

$$G = \frac{1}{\text{agm}(1, \sqrt{2})} = \frac{\Gamma(1/4)^2}{2\sqrt{2} \pi^{3/2}} \approx 0.8346 \dots$$

Extracting the gamma values from our closed form reveals that the identity is simply a rigorously scaled version of Gauss's constant:

$$G^* = 2\sqrt{\pi} \cdot G \quad (8)$$

- **The Lemniscate Constant (ϖ):** Just as π governs the perimeter of a circle, the constant $\varpi \approx 2.622$ governs the perimeter of the Bernoulli lemniscate (the ∞ -shaped curve). Because $\varpi = \pi G$, our constant scales directly with the total arc length of the

lemniscate:

$$G^* = \frac{2}{\sqrt{\pi}} \cdot \varpi \quad (9)$$

- **The Beta Function:** The Euler beta function evaluated at one-quarter is $B(1/4, 1/4) = \frac{\Gamma(1/4)^2}{\sqrt{\pi}}$. This allows us to write the identity elegantly normalized by the continuous standard normal volume:

$$G^* = \frac{1}{\sqrt{2\pi}} B(1/4, 1/4) \quad (10)$$

- **Complete Elliptic Integrals:** The complete elliptic integral of the first kind evaluated at the modulus $k = 1/\sqrt{2}$ yields $K(1/\sqrt{2}) = \frac{\Gamma(1/4)^2}{4\sqrt{\pi}}$. Thus:

$$G^* = \frac{2\sqrt{2}}{\sqrt{\pi}} K(1/\sqrt{2}) \quad (11)$$

4. The Continuous $\times$ Discrete Interpretation

Perhaps the most conceptually poetic way to look at this identity is as a direct, algebraic collision of continuous probability and discrete lattice mathematics.

Isolating the two distinct parts of the original formula reveals their profound geometric origins:

1. $\sqrt{2\pi}$ is the exact continuous volume (the normalization factor) of the one-dimensional standard Gaussian measure: $\int_{-\infty}^{\infty} e^{-x^2/2} dx$.
2. $\theta_3(e^{-\pi})^2$ is the discrete Gaussian partition function (the heat kernel trace) acting strictly over a rigid, quantized two-dimensional square integer grid, $\mathbb{Z}^2$.

Therefore, the algebraic identity can be conceptually decoupled and rewritten as the tensor product of a continuous space and a discrete boundary:

$$G^* = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \times \left(\sum_{(m,n) \in \mathbb{Z}^2} e^{-\pi(m^2+n^2)} \right) \quad (12)$$

Conclusion:

The identity accomplishes something highly remarkable. It takes the continuous probability volume of unbounded Euclidean space ($\sqrt{2\pi}$), marries it to the quantized discrete heat kernel of a two-dimensional integer lattice ($\theta_3(e^{-\pi})^2$), and flawlessly collapses the infinite complexity into a fundamental geometric constant ($\frac{\Gamma(1/4)^2}{\pi\sqrt{2}}$). In doing so, it serves as an exact analytic Rosetta Stone—permanently tethering the geometry of the discrete grid to the smooth, unbroken arithmetic-geometric mean discovered by Gauss over two centuries ago.